CN Practical 11: Locate Errors Using the Checksum Error Detection Method

Aim: To apply the checksum error detection method to a given data string and determine whether a transmission error is detected.

Theory:

Checksum is an error-detection technique used in computer networks to verify the integrity of transmitted data. The sender divides the data into equal-sized binary blocks and adds them using 1's-complement arithmetic. If an end-around carry occurs, it is added back to the sum. The 1's complement of the final sum is transmitted as the checksum along with the original data.

At the receiving end, the received data blocks and checksum are added using the same method. If the final result contains all 1s, the receiver generally assumes that no error has been detected. If the result differs, an error is detected. A checksum can detect many transmission errors, but it cannot guarantee detection of every error or reliably identify the exact erroneous bit.

Example:

Consider two 4-bit data blocks: data block 1 = 1010 and data block 2 = 1100.

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Add the blocks:
   1010
 + 1100
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  10110

Add the end-around carry:
  0110
+ 0001
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  0111

Take the 1's complement:
Sum      = 0111
Checksum = 1000
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Procedure:

1. Obtain the data string given in the problem.
2. Divide the data into equal-sized blocks.
3. Add the blocks using binary addition.
4. Add any end-around carry to the result.
5. Calculate the 1's complement of the final sum.
6. Attach the checksum to the transmitted data.
7. At the receiver, add all received data blocks and the checksum.
8. Check the final sum to determine whether an error is detected.
9. Record the result of the verification.

Observation:

The binary blocks were added using 1's-complement arithmetic, and the checksum was calculated by complementing the final sum. The receiver can use the checksum to verify the integrity of the received data. A nonmatching verification result indicates that an error has been detected, but the checksum alone does not locate the exact error bit.

Conclusion:

The checksum error detection method was applied successfully to the given data. The process of binary addition, end-around carry and complement calculation was studied. The method helps detect transmission errors and check the integrity of data.